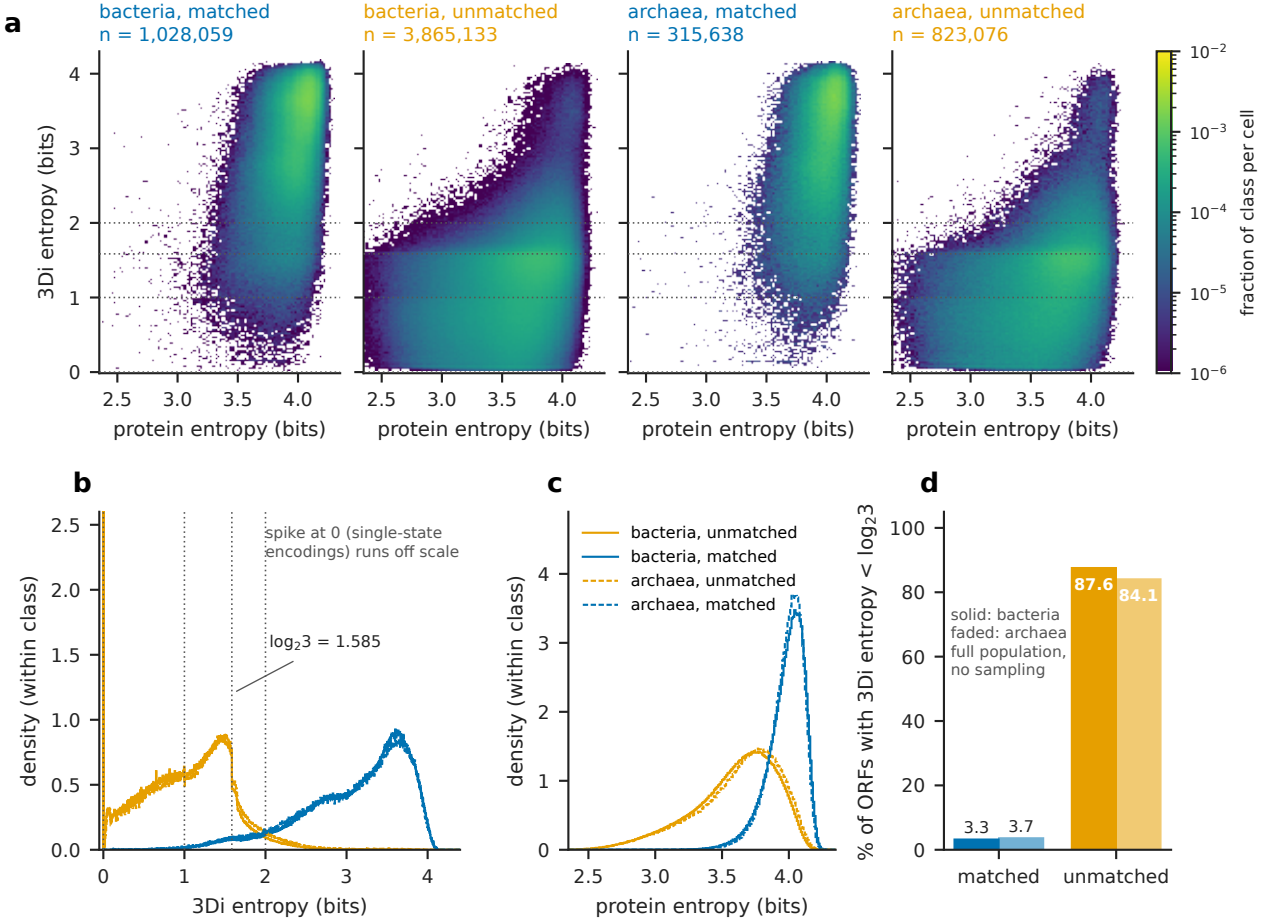

a
bacteria, matched
n = 1,028,059
bacteria, unmatched
n = 3,865,133
archaea, matched
n = 315,638
archaea, unmatched
n = 823,076
3Di entropy (bits)
protein entropy (bits)
fraction of class per cell
b
density (within class)
3Di entropy (bits)
spike at 0 (single-state encodings) runs off scale
log2 3 = 1.585
c
bacteria, unmatched
bacteria, matched
archaea, unmatched
archaea, matched
density (within class)
protein entropy (bits)
d
% of ORFs with 3Di entropy < log2 3
solid: bacteria
faded: archaea
full population, no sampling
87.6
84.1
3.3
3.7
matched
unmatched